Thi Online KTMT&HN nhóm Chẵn

Started on	Friday, 31 May 2019, 1:14 PM
State	Finished
Completed on	Friday, 31 May 2019, 2:23 PM
Time taken	1 hour 8 mins

Question 1
Complete
Marked out of 0.50

Write mask byte (in hex) to set bit 6th, 4th of a byte value with OR instruction (LSB is the 1st bit).

Answer: 3

Question 2
Complete
Marked out of 1.00

Consider the following assembly instruction sequence

```
        XOR BX, BX
        CMP DL, 5
        JLE   a_label
        CMP    DL,17h
        JGE    a_label
        MOV BX, 10h
a_label:
        INC BX
watch point:
...
```

Choose correct value of BX register at watch point for different value of DL?

DL=10 01h

DL=17h 28h

DL=0FFh 11h

DL=0Ah 6

Question 3
Complete
Marked out of 1.00

Choose correct features for SRAM and DRAM

DRAM Slower access time, cheaper cost per bit, can manufacture with larger size

SRAM Faster access time, cheaper cost per bit, can manufacture with larger size

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Question 4
Complete
Marked out of 1.00

Consider a 16-bit microprocessor, with a 16-bit external data bus, drivenby an 10-MHz input clock. Assume that this microprocessor has a bus cycle whose minimum duration equals four input clock cycles. What is the maximum data transfer rate across the bus that this microprocessor can sustain?

Select one:

- 1 MB/s
- 10 MB/s
- 5 MB/s
- 4 MB/s

Question 5
Complete
Marked out of 1.00

the memory stack area of a program shown in figure

Address	1D50	1D51	1D52	1D53
Value	AF	90	71	DA

The value of SP register is 1D50. What isthe value of SP follows the execution of **PUSH SI**

Answer: 1D4F

Question 6
Not answered
Marked out of 1.50

A benchmark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	Cycles per Instruction
Integer arithmetic	45,000	1
Data transfer	32,000	2
Floating point	15,000	2
Control transfer	8000	2

Calculate MIPS rate for this program

Given:

$$\text{MIPS rate} = \frac{I_c}{T \times 10^6} = \frac{f}{CPI \times 10^6}$$

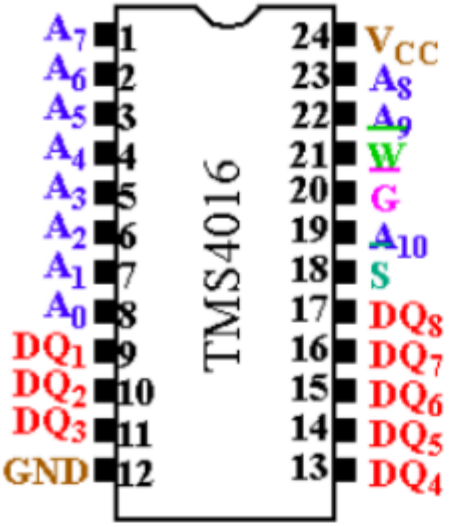
$$CPI = \frac{\sum_{i=1}^{n}(CPI_i \times I_i)}{I_c}$$

Answer:

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Question 7
Complete
Marked out of 1.00

Choose the correct structure ofmemorychipasshownbelow



Note:
DQ: Data pinout

Select one:

- DRAM 2Kx8-bit
- SRAM 2Kx8-bit
- SRAM 1Kx16-bit
- DRAM 1Kx16-bit

Question 8
Complete
Marked out of 1.00

What is the correct value of SI, AL (in hex) at watch point:

```
01:             MOV SI, 300h
02:             MOV AL, 10h
03:             MOV CX, 7
04: Loop_label:
05:             MOV [SI], AL
06:             ADD AL,10h
07:             INC SI
08:             LOOP Loop_label
```

watch point:

AL = 307h

SI 308h

Question 9
Complete
Marked out of 1.00

The principle of cache memory relies on key features: locality of reference which involves spatial and temporal locality. Match the definition to keywords on the left

Spatial locality — the tendency of execution to involve a number of memory locations that are clustered

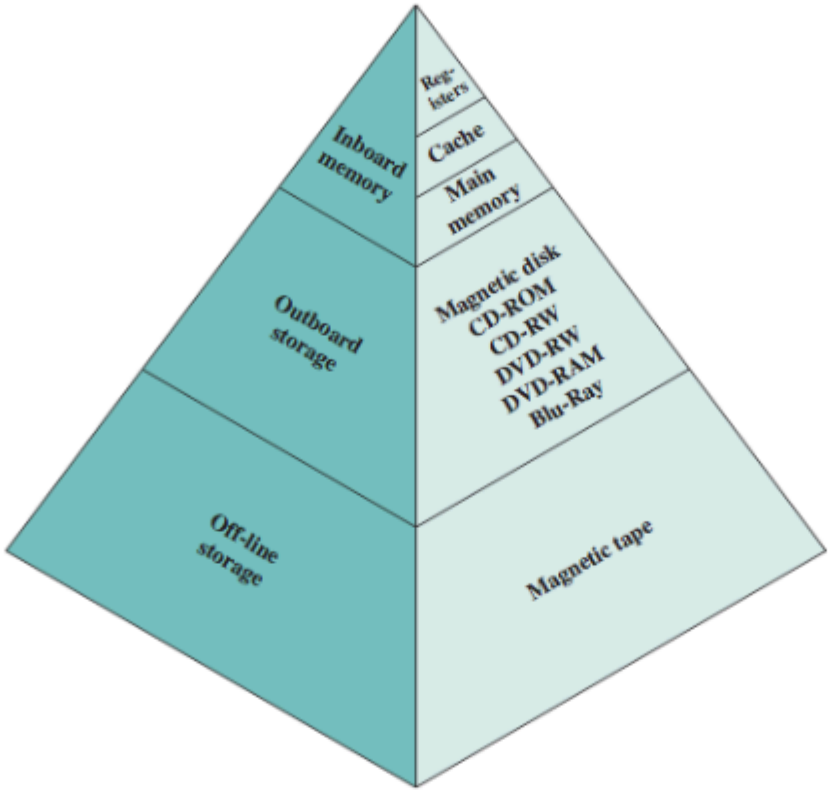
Temporal locality — the tendency for a processor to access memory locations that have been used recently

tendency to use large cache and prefetch mechanism

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Question 10
Complete
Marked out of 1.00

For memory hierarchy below, whichrelationshipholdwhenmovingdownward



Select one or more:

- [ ] Decreasing cost per bit
- [ ] Decreasing frequency of access by the processor
- [ ] Increasing access time
- [ ] Increasing capacity
- [x] the processor accesses more often

Question 11
Complete
Marked out of 1.00

Given 8-bit floating-point binary format:
1 (sign) + 3 (exponent) + 4 (mantissa)
Convert the 8-bit floating point number 68 (in hex) to decimal.

Answer: 12 104

Question 12
Complete
Marked out of 1.00

In computer organization, the CPU transfer rate is much higher than that of memory. It is easy to match performance of these components by:

Select one:

- (•) increase the bus speed
- ( ) increase I/O speed
- ( ) producing faster memory module
- ( ) Introducing cache memory

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Question 13
Complete
Marked out of 0.50

The instruction that loads the AHregisterwiththelowerbyteoftheflagregister is

Select one:

- SAHF
- AH
- LAHF
- PUSHF

Question 14
Complete
Marked out of 1.50

Which are correct about 32 bit index registers of IA-32 processors:

Select one or more:

- ESH,EDH: 16 bit pointers to higher memory above 1M
- SI: 16 bit pointer to source memory in data movement instructions
- ESI: 32 bit pointer to source memory in data movement instructions
- EDI: 32 bit pointer to destination memory in data movement instructions
- DI: 16 bit pointer to destination memory in data movement instructions

Question 15
Complete
Marked out of 1.00

The following sequence of instructions are executed. What is the correct value of flag bits at watch point?

```
MOV AL, 0F
ADD AL, F1
```

watch point:

Zero flag (OF) = set

Carry flag (CF) = set

Question 16
Complete
Marked out of 1.20

What is the correct sequence of instruction cycle?

Step 6 Store result

Step 1 Fetch opcode

Step 2 Calculate operand address

Step 3 Fetch operand

Step 5 Decode

Step 4 Execution

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Question 17
Complete
Marked out of 0.50

To encrypt a byte value, use _____instruction.

Select one:
- AND
- NOT
- OR
- XOR

Question 18
Complete
Marked out of 0.50

Which of the following instructions are not valid?

Select one or more:
- [x] MOV AX, SI
- [ ] MOV AX, [BP+2]
- [ ] MOV DS, B800h
- [x] MOV SP, SS:[SI+2]

Question 19
Complete
Marked out of 0.50

8088 is 16 bit processor, the maximum addressable memory is:

Select one:
- 640K
- 64M
- 640M
- 1024K

Question 20
Complete
Marked out of 1.00

Select correct definition of seek time, rotational delay, access time, transfer time for hard drives with moveable-head system:

rotational delay: access time + settle time
seek time: time for the head to settle at the request track
access time: transfer time

Question 21
Complete
Marked out of 1.00

Which of the following instructions are not legal addressing?

Select one or more:
- [x] MOV AX, [SP+1]
- [ ] MOV AX, [DI]
- [ ] MOV CX, [SI]
- [ ] MOV AX, [BX+SP]

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Question 22
Complete
Marked out of 1.00

A memory chip has 12 addresspins,determinethemaximummemorywordsof this chip?

Select one:

- 2048
- 4096
- 2048K
- 4000

Question 23
Complete
Marked out of 1.00

Convert 0.1015625 to IEEE 32-bit floating point format (1 sign+ 8 exponent + 23 mantissa)

Answer: 3DD00000

Question 24
Complete
Marked out of 1.00

Structural components of computer include:

Select one or more:

- [x] Memory
- [x] Central processing unit
- [x] I/O
- [ ] DMA
- [x] Interrupt
- [ ] System interconnection

Question 25
Complete
Marked out of 1.00

Select the correct sequence of instructions to compute -1024/128 (all values are in hex).

Step 1: MOV AX,FC00
Step 2: IDIV CX
Step 3: MOV CX,80
Step 4: CWD

Question 26
Complete
Marked out of 0.50

Sign-extend number 1011 0101 (8-bit binary) to 16-bit

Answer: B5

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Question 27
Complete
Marked out of 1.00

Part of computer memory is showninfigure

Address	1D48	1D49	1D4A	1D4B	1D4C	1D4D	1D4E	1D4F
Value	03	7F	F5	2D	5A	12	7B	C0

What is the value of AX register after instruction **MOV AX, [1D4B]** executed

Answer: 5A2D

Question 28
Complete
Marked out of 2.00

Choose correct RAID volume definitions for a request 2T storage.

Spanned Volume	2T HDD + more HDDs to extend storage, no fault tolerance, data lost when one HDD fails
RAID5 Volume	At least 3 x 2T HDDs, fault-tolerance, no data lost, no down-time
RAID 1 - Mirror volume	2 x 2T HDDs are needed, no data lost when the primary storage fails
RAID 0 - Striped volume	2 x 1T HDDs are needed, enhance data transfer, no fault tolerance, data lost when one HDD fails

Question 29
Complete
Marked out of 1.00

Select correct level for contemporary computer multilevel machine

Level 1	Operating system level
Layer 4	Microarchitecture level
Level 3	Digital logic level
Level 2	Digital logic level
Level 5	Internet
Level 0	Assembly Language level
Level 6	High level programming language

Question 30
Complete
Marked out of 0.50

the instruction, CMP to compare source and destination operands by _____

Select one:

- dividing
- comparing
- adding
- subtracting

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Question 31
Complete
Marked out of 0.50

Write mask byte (in hex) to clearbit2nd,3rd,5thofabytevaluewithAND instruction (LSB is 1st bit).

Answer: 5

Question 32
Complete
Marked out of 1.00

To evaluate processor performance, the following indicators and formulas are used:

Cycles per instruction $$CPI = \frac{\sum_{i=1}^{n}(CPI_i \times I_i)}{I_c}$$

Time to execute a program $$T = I_c \times CPI \times \tau$$

Or $$T = I_c \times [p + (m \times k)] \times \tau$$

In which:
p: the number of processor cycles needed to decode and execute the instruction
m: the number of memory references needed
k: the ratio between memory cycle time and processor cycle time
τ: cycle time = *1/f*

Which of the following system attributes affects cycle time t

Select one or more:

- [x] Compiler technology
- [ ] Cache and memory hierarchy
- [ ] Processor implementation
- [x] Instruction set architecture

Question 33
Not answered
Marked out of 1.00

Convert the 32-bit floating point number C4361000 (in hex) to decimal.

Answer:

Question 34
Complete
Marked out of 1.00

Which is correct about dual-layer DVD?

Select one:

- contains layers on both sides of the disk for writing data to
- the same as double-sided DVD
- contains two layers on a single side for writing data to
- DVD drives has double laser head for reading from or writing to this disk

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Question 35
Complete
Marked out of 1.00

The following sequence of instructionsareexecuted.Whatisthecorrectvalue of AX, CX, DX at watch point?

```
MOV AX,0020
MOV CX,0010
MUL CL
```

watch point:

AX = 0200 ▼

DX 0000 ▼

CX = 0010 ▼

Question 36
Complete
Marked out of 0.50

In multiplication instruction, the upper half of the result is nonzero implies which state of Carry flag and Overflow flag?

Select one or more:

- [x] CF=0
- [x] OF=1
- [ ] OF=0
- [ ] CF=1

Question 37
Complete
Marked out of 1.00

Which statements are correct for HDDs?

Select one or more:

- [x] a. Head, Track, Sector are key parameters for access data on hard disk
- [ ] b. Head, Track, Cylinder are key parameters for access data on hard disk
- [x] c. Bits are stored on tracks
- [ ] d. Bits are store randomly on disk surfaces

Question 38
Complete
Marked out of 1.00

Consider a magnetic disk drive with 8 surfaces, 512 tracks per surface, and 64 sectors per track. Sector size is 1 kB. What is the disk capacity

Answer: 16.3 KB ▼

Question 39
Complete
Marked out of 0.50

Which are the correct inputs for XLAT instruction

Select one or more:

- [ ] look-up index must be loaded into AL
- [x] look-up index must be loaded into DL
- [ ] DS:[BX] pointed to look-up table DS:
- [x] [SI] pointed to look-up table

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Question 40
answered
Marked out of 1.00

A benchmark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count:

Instruction Type	Instruction Count	Cycles per Instruction
Integer arithmetic	45,000	1
Data transfer	32,000	2
Floating point	15,000	2
Control transfer	8000	2

Calculate the execution time for this program.
Given:

$$T = I_c \times CPI \times \tau$$

$$CPI = \frac{\sum_{i=1}^{n}(CPI_i \times I_i)}{I_c}$$

Answer:

Question 41
Complete
Marked out of 0.50

In multiplication instruction, when the source operand is 8 bit, _____ will be multiplied with source.

Select one:

- BX
- AL
- AX
- Whatever general purpose register

Question 42
Complete
Marked out of 0.50

Which are the correct actions for LODSW string operation if DF is reset (=0)

Select one or more:

- Load 16-bit value at memory location pointed by DS:[SI] into AX
- decrease DI by 2
- Load 16-bit value at memory location pointed by ES:[DI] into AX
- increase SI by 2

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Question 43
Complete
Marked out of 1.50

A system programmer needs to compute 449/2+358/4(decimal).Instructhim to code in debug (number must be in hex) with the least number of instruction counts.

Step 1: MOV BX,AX

Step 2: ADD AX,BX

Step 3: MOV CL,2

Step 4: ADD AX,166

Step 5: SHL AX,CL

Step 6: SHR AX,CL

Question 44
Complete
Marked out of 1.00

For better speed, in CPU design, engineers make use of the following techniques:

Select one or more:

- [x] Speculative execution
- [x] Branch prediction
- [x] Pipelining
- [ ] Faster CPU internal bus

Question 45
Complete
Marked out of 1.00

Which ones are not correct for static RAM?

Select one or more:

- [x] Cheaper than dynamic RAM because simpler chip controller
- [x] Cost per bit is lower than dynamic RAM
- [ ] faster than dynamic RAM because they are made from capacitor
- [ ] Cost per bit is higher than dynamic RAM

◄ Announcements

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